BASIC ALU IMPLEMENTATION

Internship Project Report

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Project Name: Basic ALU Implementation

Duration: 4 Weeks

Software Used: Xilinx Vivado, Verilog HDL

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1. Introduction

An Arithmetic Logic Unit (ALU) is one of the most important components of a digital system. It is responsible for performing arithmetic and logical operations on binary data. Every processor and microcontroller contains an ALU that executes mathematical calculations and logical comparisons.

In this project, a **4-bit Basic ALU** was designed using **Verilog Hardware Description Language (HDL)** and simulated using **Xilinx Vivado**. The ALU performs various arithmetic and logical operations depending on the value of the selection signal (Sel).

The project demonstrates the implementation of digital logic using Verilog and verifies its functionality through simulation.

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2. Objective

The objectives of this project are:

- To design a 4-bit Arithmetic Logic Unit using Verilog HDL.
- To implement arithmetic and logical operations.
- To simulate the design using Xilinx Vivado.
- To verify the correctness of the output using a testbench.

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3. Software Used

- Xilinx Vivado
- Verilog HDL
- Windows Operating System

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4. Theory

An Arithmetic Logic Unit (ALU) is a combinational digital circuit that performs different arithmetic and logical operations. The required operation is selected using a control signal known as the selection input (Sel).

The ALU designed in this project accepts two 4-bit inputs, A and B, along with a 3-bit selection input. Depending on the selection value, the ALU performs one of the supported operations.

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5. Operations Performed

Sel	Operation
000	Addition
001	Subtraction
010	AND
011	OR
100	XOR
101	NOT
110	Left Shift
111	Right Shift

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6. Inputs and Outputs

Inputs

- A (4-bit input)
- B (4-bit input)
- Sel (3-bit selection signal)

Outputs

- Result (4-bit output)
- Carry (1-bit output)

Working Principle

The ALU receives two binary inputs (A and B) and a 3-bit selection signal. Based on the value of Sel, the ALU performs the corresponding arithmetic or logical operation. The result is displayed at the output. During addition, the Carry output indicates whether an overflow has occurred.

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7. Verilog Design

The project contains two Verilog files:

alu.v

This is the main design module that implements the ALU. A case statement is used to select different arithmetic and logical operations according to the selection input.

alu_tb.v

This is the testbench module. It applies different input values and selection signals to verify that the ALU performs all operations correctly.

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8. Simulation

The design was simulated using the Xilinx Vivado Behavioral Simulator.

The following input values were used during simulation:

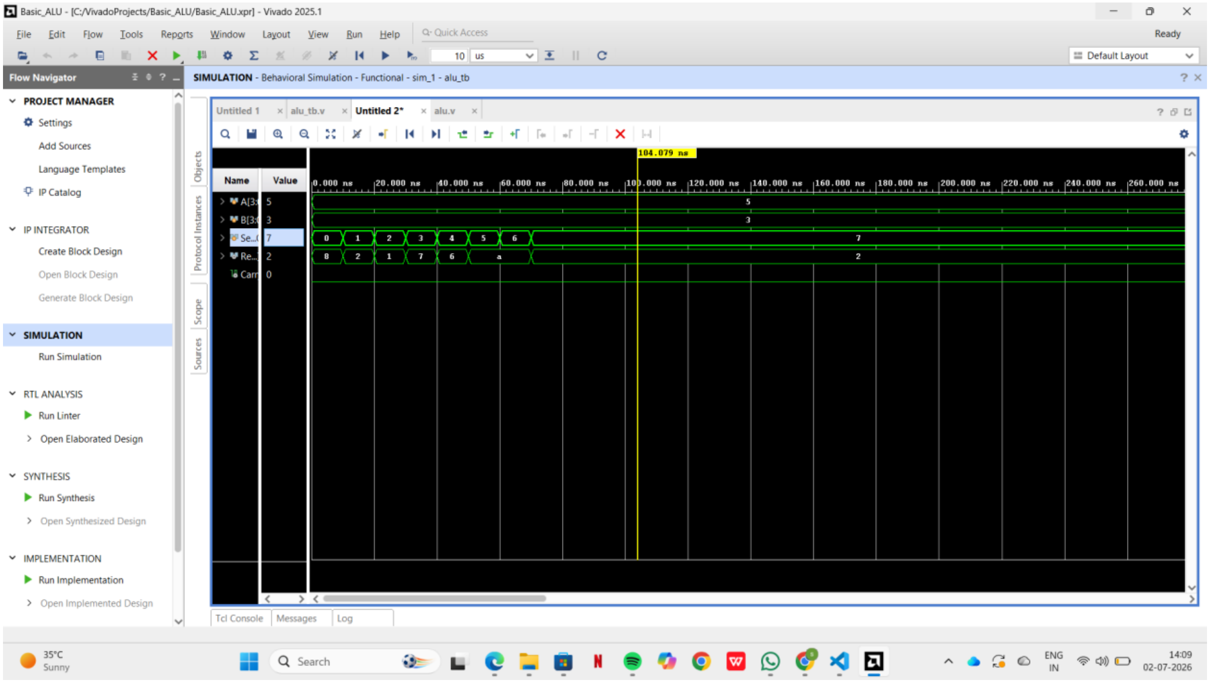
Input A = 0101 (5)

Input B = 0011 (3)

Different values of the selection signal (Sel) were applied sequentially to test each ALU operation.

Simulation Results

Operation	Output
Addition	8
Subtraction	2
AND	1
OR	7
XOR	6
NOT	1010
Left Shift	1010
Right Shift	0010



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9. Advantages

- Simple and easy to understand.
- Performs multiple arithmetic and logical operations.
- Can be extended to higher bit-width ALUs.
- Useful for learning digital design concepts.
- Efficient implementation using Verilog HDL.

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10. Applications

The Arithmetic Logic Unit is widely used in:

- Microprocessors
- Embedded Systems
- Digital Signal Processing
- Computer Architecture
- Arithmetic Processing Units
- FPGA-based Digital Systems

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11. Conclusion

The 4-bit Basic ALU was successfully designed and implemented using Verilog HDL. The design was simulated using Xilinx Vivado, and all arithmetic and logical operations produced the expected results. The project demonstrates the basic functionality of an Arithmetic Logic Unit and provides a strong foundation for understanding digital system design using hardware description languages.

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12. References

1. Xilinx Vivado Design Suite User Guide.
2. IEEE Standard for Verilog Hardware Description Language.
3. Digital Design by M. Morris Mano.
4. Verilog HDL by Samir Palnitkar.
